

Division Expressions with Fractions and Whole Numbers

Lesson 6-1

Name: _____

Date: _____

Class: _____

EVIDENCE LAB MISSION

| The Fraction Vault

You are the lead forensic analyst at the Reyes Detective Agency. A 3-foot strip of evidence tape must be cut into exact $\frac{1}{4}$ -foot sections before it can be logged — and the vault that holds the case file only opens once you know how many pieces that makes. Interpret fraction division and crack the case.



TODAY'S OBJECTIVES

Content Objective: I can divide a whole number by a unit fraction and divide a fraction by a whole number by writing the whole number over 1 and using Keep, Change, Flip.

Language Objective: I can explain each step of Keep, Change, Flip using the words reciprocal, divisor, and quotient.



See **Learn It** for the explanation and worked examples, and the **Vocab** tab for the words. These are your notes to fill in and keep.



Fill in each blank as we go. Use the Word Bank to help you.



WORD BANK — FILL EACH BLANK WITH THE BEST WORD

Reciprocal

Quotient

Dividend

Divisor

Unit fraction

Multiplicative inverse

Tape diagram



Tap any word to see its meaning and a picture.

Level 3: try the blanks from memory first, then check the bank.

1

Flipping the numerator and denominator gives the

.

2

The answer to a division problem is the

.

3

In $6 \div \frac{1}{2}$, the number 6 being divided is the

.

4

In $6 \div \frac{1}{2}$, the number $\frac{1}{2}$ we divide by is the

.

5

A fraction with a numerator of 1, like $\frac{1}{5}$, is a

.

6

Two numbers whose product is 1 are

of each other.

7

A bar cut into equal parts to show sharing is a .

Write About the Math

How does interpreting fraction division help the detective?

¿Cómo ayuda interpretar la división de fracciones al detective?

Say your answer to a partner first. Then write it, using at least two of these words:

☐ **Reciprocal**
Recíproco

☐ **Quotient**
Cociente

☐ **Dividend**
Dividendo

☐ **Divisor**
Divisor

☐ **Unit fraction**
Fracción unitaria

Model: *There will be more than 3 sections because each piece is smaller than a whole foot.* — Students reason that dividing by a number less than 1 gives a quotient greater than the dividend, so there are more than 3 sections.

Start

Complete the frames. Use the word bank.

Completa las oraciones. Usa el banco de palabras.

- **The detective can mark ____ sections because $6 \div 1/3 =$ ____.**
- **My answer is ____ because ____.**

Mi respuesta es ____ porque ____.

Build

Write two connected sentences. Use because, so, or but.

Escribe dos oraciones conectadas. Usa porque, entonces o pero.

- **My claim is ____.**
Mi afirmación es ____.
- **My evidence is ____, so ____.**
Mi evidencia es ____, entonces ____.

Explain

Write a claim, give math evidence, and explain your reasoning.

Escribe una afirmación, da evidencia matemática y explica tu razonamiento.

☐ I answered the question.

☐ I used math evidence: numbers, an equation, or a model.

☐ I used at least two math words.